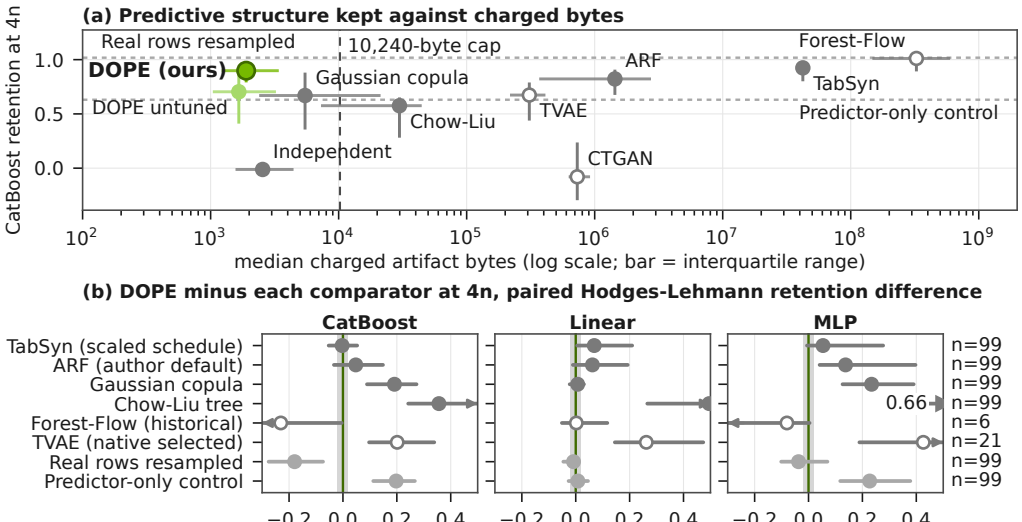
(a) Predictive structure kept against charged bytes
CatBoost retention at 4n
1.0
0.5
0.0
Real rows resampled
DOPE (ours)
DOPE untuned
10,240-byte cap
Gaussian copula
Chow-Liu
Independent
ARF
TVAE
CTGAN
Forest-Flow
TabSyn
Predictor-only control
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
median charged artifact bytes (log scale; bar = interquartile range)
(b) DOPE minus each comparator at 4n, paired Hodges-Lehmann retention difference
CatBoost
Linear
MLP
TabSyn (scaled schedule)
ARF (author default)
Gaussian copula
Chow-Liu tree
Forest-Flow (historical)
TVAE (native selected)
Real rows resampled
Predictor-only control
n=99
n=99
n=99
0.66
n=99
n=6
n=21
n=99
n=99
−0.2
0.0
0.2
0.4